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pH-dependent emulsifying properties of pea [*Pisum sativum* (L.)] proteins

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ABSTRACT

Emulsifying properties of two partially purified legumin and vicilin (PL and PV) and protein isolate (PPI) from dry pea seeds at various pH values (3.0, 5.0, 7.0 and 9.0) were investigated. The tested emulsion characteristics included droplet size, flocculation and coalescence indices (FI and CI), creaming index, as well as interfacial protein adsorption. Some physicochemical properties of these proteins, e.g., free sulfhydryl and disulfide bond contents, protein solubility (PS), surface hydrophobicity (H_o) and thermal stability (and denaturation), were also characterized. The results indicated that emulsifying ability and emulsion stability of various pea proteins considerably varied with the preparation process, protein composition and pH. Overall, all the pea proteins exhibited least emulsifying ability at pH 5.0 (around isoelectric point), and concomitantly, the resultant emulsions were most unstable against coalescence and creaming. The emulsifying ability of these proteins at pH 3.0 was generally better than that at neutral or alkali pH values, and among all the three proteins, PL exhibited highest emulsifying ability at this pH. The flocculated state and size of droplets in fresh emulsions did not directly affect stability of these emulsions against flocculation and coalescence (upon 24 h of storage), and even creaming (up to 7 days). Interestingly, the PL and PV exhibited much better creaming stability than PPI, at pH deviating from the *p*I. The emulsifying properties of these proteins were not only related to their PS and H_o, but also associated with the protein adsorption and nature (e.g., viscoelasticity) of interfacial protein films. These results can greatly extend the knowledge for understanding the emulsifying properties of pea proteins, especially the pH dependence of emulsion characteristics.

1. Introduction

There is increased interest in utilizing pea [*Pisum sativum* L.] protein as an alternative for soy protein that has a dominating advantage in the market. Pea is one of the valued crops in the world market (Adebiyi & Aluko, 2011; Tian, Kyle, & Small, 1999); like other legume seeds, pea seed is characteristically rich in proteins (18–30%) with a well-balanced amino acid profile, especially a high content in lysine (Schneider & Lacampagne, 2000). In addition to providing amino acid nutrition, the ultimate success of utilizing pea protein as a promising food ingredient and an alternative to soy proteins depends largely on its functional properties, including solubility, viscosity, water- and oil-binding property, gelation, foaming and emulsifying properties. To date, pea protein products are very limited in food applications, though pea protein isolate (PPI) has been commercially available in the market. One of the major limitations is that the functionality of pea protein products (especially PPI) as a food ingredient is relatively weak, as compared with soy protein isolate (SPI) (Shand, Ya, Pietrasik, & Wanasundara, 2007; Sun & Arntfield, 2011a). However, it would be more probably due to lack of knowledge about structural and functional properties of pea proteins (Adebiyi & Aluko, 2011; Aluko, Mofolasayo, & Watts, 2009; Karaca, Low, & Nickerson, 2011).

The majority of pea proteins are storage proteins; legumin and vicilin (usually classified as 11S and 7S globulins) are the two major storage proteins representing 65–80% of the total buffer-extractable protein (Koyoro & Powers, 1987). The 11S legumin has regularly a hexameric quaternary structure (six subunits), with one subunit consisting of an acidic polypeptide and a basic one (of about 40 and 21 kDa, respectively) that are linked together by a disulfide bridge. The 7S vicilin is a trimer of 50-kDa subunits and constitutes about 35% of the total protein content of the seeds. Some of the 50-kDa subunits are nicked by proteolysis soon after biosynthesis, but the resultant peptides (ranging from 12.5 to

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33 kDa) remain associated with intact subunits in the native 150 kDa oligomer (Pedrosa, Trisiuzzi, & Ferreira, 1997).

Emulsifying property is one of the most important functionalities for proteins as food ingredients in a wide range of applications. As compared with gelation properties of pea proteins that have been relatively widely studied (O'Kane, Happe, Vereijken, Gruppen, & van Boekel, 2004a; O'Kane, Vereijken, Gruppen, & van Boekel, 2005; Shand et al., 2007; Sun & Arntfield, 2010, 2011a, 2011b, 2012), very little information has been available addressing their emulsifying properties. In an early investigation, Koyoro and Powers (1987) investigated the emulsion capacity and stability of salt-extracted globulins fractions (legumin and vicilin) at pH 3.0–7.0, and found that vicilin exhibited more emulsion stability but less emulsion capacity than legumin. This has been attributed to the fact that vicilin, because of its lower molecular weight and more flexible tertiary structure, is a more effective surface-active globulin than legumin (Dagorn-Scaviner, Gueguen, & Lefebvre, 1987). Previous research shows that pea protein products have emulsifying properties comparable to SPI, possibly due to their better solubility (Aluko et al., 2009; Swanson, 1990), but it still needs to be further confirmed. Herein, one point to note is that the emulsifying properties of pea proteins highly vary with their protein composition, e.g. vicilin/legumin ratio, with better emulsifying properties observed for vicilin or vicilin-rich proteins. Rangel, Domont, Pedrosa, and Ferreira (2003) confirmed that at neutral pH, pea purified vicilin exhibited a higher emulsifying capacity (as evidenced by higher emulsifying activity index, EAI) than casein and bovine serum albumin, two well-recognized proteins with excellent emulsifying properties. This is further confirmed by the observation of Barac et al. (2010) that the difference in vicilin to legumin ratio between PPIs from different pea genotypes considerably affects their emulsifying properties.

Another interesting point is noteworthy that the emulsifying properties of pea proteins are highly dependent on the preparation process or method. It is generally recognized that salt-extracted protein isolates exhibited better functionalities than isoelectric (alkali extraction and acid precipitation) isolates (Swanson, 1990). However, it is not always the case. For example, in a recent work Karaca et al. (2011) showed that although the emulsion capacity or activity of salt-extracted PPI was similar to that of isoelectric-precipitated PPI, but the emulsion stability was much less in terms of mean droplet diameter and creaming index. Furthermore, some investigations reported that pea protein products obtained by ultrafiltration exhibited better functionalities (including emulsifying properties) than those by isoelectric precipitation (Boye et al., 2010; Fuhrmeister & Meuser, 2003; Taherian et al., 2011).

The emulsifying properties of pea proteins are also affected by environmental parameters, e.g., pH. In general, pea protein products exhibit better emulsifying properties at a pH deviating from isoelectric point (*p*I) than at a pH around *p*I (Adebiyi & Aluko, 2011; Fuhrmeister & Meuser, 2003). This is consistent with the fact that good solubility is a prerequisite for the emulsifying properties of proteins (Damodaran, 1996). However, Aluko et al. (2009) indicated that variation in pH (pH 3.0–7.0) did not result in a significant change in emulsion oil droplet size of a commercial PPI, where the authors attributed this to higher concentrations of proteins and lower homogenization pressure, applied in their work. Gharsallaoui, Cases, Chambin, and Saurel (2009) compared the surface adsorption behavior and emulsifying characteristics of PPI at pH 2.4 and 7.0, and found that although the adsorption of PPI at the interface is slower at acidic conditions, but the formed emulsions are more stable against creaming at acidic pH. This interesting finding suggests a close relationship between the interfacial and emulsifying properties of PPI as can be tuned by varying pH.

Although all the mentioned above works had generally indicated that pea proteins exhibit good potential to act as an emulsifying protein agent, especially at acidic conditions, and in this aspect, are even comparable to soy proteins, there are still a number of issues that need to be addressed, such as the pH dependence of physicochemical and emulsifying properties of pea vicilin and legumin. The main objective of the work thus aims to investigate the emulsifying properties of pea purified globulins (vicilin and legumin) and protein isolate at pH values ranging from 3.0 to 9.0. The emulsifying properties (including emulsifying ability and emulsion stability) were evaluated using droplet-size analysis and visual observation (especially creaming stability). Some physicochemical properties such as free sulfhydryl and disulfide bond contents, surface hydrophobicity and protein solubility were also evaluated.

2. Materials and methods

2.1. Materials

Dry pea [*P. sativum* (L.)] seeds were purchased from a local supermarket in Guangzhou (China). The seeds were soaked in deionized water for 12 h at 4 °C and dehulled manually. The dehulled seeds were freeze-dried, ground and defatted by Soxhlet extraction with hexane to produce the defatted flour. 5,5′-dithio-bis 2-nitrobenzoic acid (DTNB), 2,4,6-trinitrophene sulphonic acid (TNBS) and 1, 8-anilinonaphthalenesulfonate (ANS^-) reagents were purchased from Sigma–Aldrich (St. Louis, MO, USA). Bovine serum albumin (BSA) was obtained from Fitzgerald Industries International Inc. (Concord, MA, USA). All other chemicals used were of analytical grade.

2.2. Preparation of pea protein isolate (PPI)

PPI was prepared according to an alkali extraction/acid precipitation process described by Sumner, Nielsen, and Youngs (1981), with a few modifications. Dehulled and freeze-dried pea seed flour (100 g) was dispersed in distilled water at a solid/solvent ratio of 1:10 (w/v) and stirred for 1 h to obtain a uniform dispersion. The resultant dispersion was adjusted with 2 M NaOH to pH 9.0, and further stirred for another 1 h and then centrifuged (8000 *g*, 30 min) to remove the insoluble material. The pH of the obtained supernatant was adjusted to 4.5 with 2 M HCl, and stored for 2 h at 4 °C and last centrifuged (5000 *g*, 20 min). The obtained precipitate was washed with distilled water (4 °C) and re-dispersed in distilled water with pH adjusted to 7.5. The dispersion was further centrifuged at the same conditions, and the last supernatant was dialyzed against distilled water (at 4 °C for 48 h) and lyophilized to produce the PPI sample.

2.3. Preparation of partially purified legumin and vicilin (PL and PV)

The PL and PV were prepared according to the method described by Klassen and Nickerson (2012), with a few modifications. The whole schematic flow for the preparation is depicted in Fig. 1. In brief, 200 g of dehulled and freeze-dried pea seed flour was dissolved in 50 mM a K_2PO_4 buffer (pH 7.2) with 0.5 M NaCl at a solid/solvent ratio of 1:10 (w/v), and then stirred continuously using a magnetic stirrer for 1 h at room temperature. The dispersion was centrifuged at 8000 *g* for 20 min at 4 °C. The resultant supernatant was then diluted using 5 volumes of cold (4 °C) deionized water, and adjusted to pH 4.5 using 2 M HCl and then left to stand at 4 °C for around 2 h to facilitate the settling of salt-soluble proteins. The supernatant was decanted off and the remaining pellet was collected by centrifugation (8000 *g*, 20 min). The pellet was resuspended within the same buffer at a ratio of 5 mL per g of pellet, and then allowed to stir continuously for 1 h at 4 °C. This last extraction was repeated, and subsequent supernatants pooled.

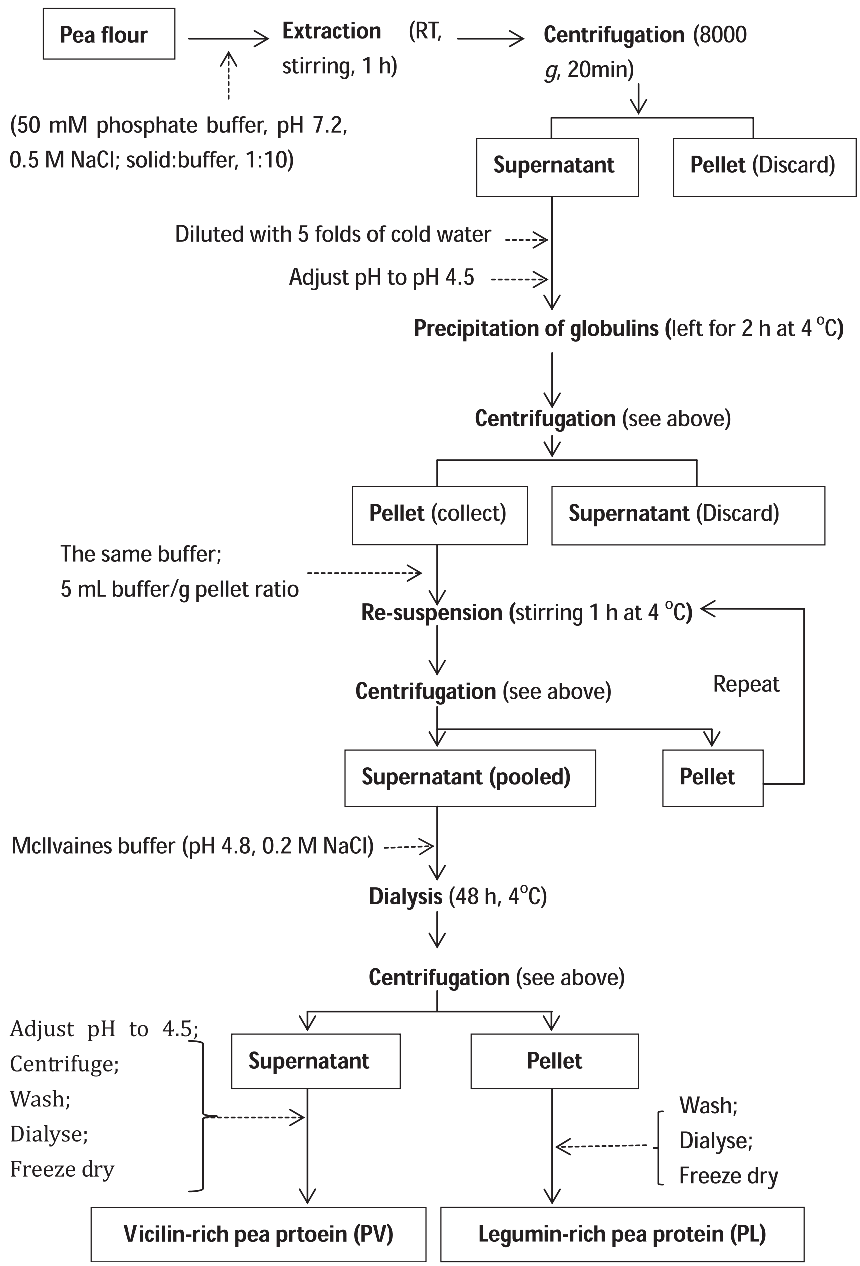

Fig. 1. Schematic flow illustration for the preparation partially purified legumin and vicilin-rich pea proteins (denoted as PL and PV, respectively).

Approximately half of the pooled supernatant was dialyzed against a McIlvaines buffer (0.2 M Na_2HPO_4 and 0.1 M citric acid, pH 4.8) containing 0.2 M NaCl, with the buffer being changed twice using a volume ratio of 10 mL buffer per 1 mL of supernatant. The precipitate collected by centrifugation (with the supernatant saved for the later use) was re-solubilized in the phosphate buffer, and dialyzed against deionized water at 4 °C and then lyophilized to produce the legumin-rich pea protein (denoted as PL). The last supernatant was adjusted to pH 4.5 using 1 M HCl to precipitate the vicilin component. The resultant precipitate was re-solubilized, dialyzed and lyophilized as the previously described to produce the vicilin-rich pea protein (denoted as PV).

2.4. SDS-PAGE

SDS-PAGE was performed on a discontinuous buffered system according to Laemmli's method (Laemmli, 1970) that utilized 14% separating gel (pH 8.8) and 5% stacking gel (pH 6.8). The protein samples (8%, w/v) were fully solubilized in the electrophoresis buffer with (reducing) or without (nonreducing) 5% (w/v) β-mercaptoethanol. After being heated in boiling water for 5 min and then centrifugation (10,000 *g*, 10 min), these samples (7.5 μL each sample) were loaded on the gels. The gels was stained with 0.25% Coomassie brilliant blue (R-250) in 10% acetic acid and 45% methanol, and destained in methanol-water solution containing 10% (v/v) acetic acid (methanol:acetic acid:water = 1:1:8, v/v/v).

2.5. Determination of free sulfhydryl group (SH) and disulfide bond (SS) contents

The free SH (including total and exposed) and SS contents of protein samples were determined by the method of Beveridge, Yoma, and Nakai (1974), as described in our previous work (Tang & Sun, 2010).

2.6. Surface hydrophobicity (H_o)

H_o of proteins was determined with the fluorescence probe ANS$^-$ according to the method of Haskard and Li-Chan (1998). Serial dilutions in 0.01 M phosphate buffer (pH 7.0) were prepared with the protein samples (stock solutions; 1.5%, w/v) to a final protein concentration of 0.004–0.02% (w/v). ANS$^-$ stock solution (8.0 mM) was also prepared in the same buffer. Twenty μL of ANS$^-$ solution was added to 4 mL of each dilution and fluorescence intensity (FI) of the mixture was measured at 370 nm (excitation) and 470 nm (emission) using F4500 fluorescence-spectrophotometer (Hitachi Co., Japan). Initial slope of FI vs protein concentration (mg/mL) plot (calculated by linear regression analysis) was used as an index of H_o.

2.7. Differential scanning calorimetry (DSC)

Thermal transition of various protein samples was examined using a TA Q100-DSC thermal analyzer (TA Instruments, New Castle, Delaware 19720 USA). Approximately 2.0 mg of samples were weighed into aluminum liquid pans (Dupont), and 10 μL of 0.01 M phosphate buffer (pH 7.0) was added. Pans were hermetically sealed and heated from 20 to 120 °C at a rate of 5 °C/min. A sealed empty pan was used as a reference. Onset temperature (T_m), peak transition or denaturation temperature (T_d), and enthalpy change of endotherm (ΔH) were computed from the thermograms by the Universal Analysis 2000, Version 4.1D (TA Instruments-Waters LLC). All experiments were conducted in triplicate. The sealed pans containing protein samples and buffers were equilibrated at 25 °C for more than 6 h.

2.8. Protein solubility (PS)

An aqueous solution (approximately 1.0%, w/v) of protein samples in deionized water was stirred magnetically for 30 min, and adjusted to desired pH values in the range 2.0–10.0 (or pH 12.0) using either 0.5 M HCl or 0.5 M NaOH. After 30 min of stirring, it was centrifuged at 8000 *g* for 20 min at 20 °C in a CR22G centrifuge (Hitachi Co., Japan). After an appropriate dilution with deionized water with the same pH, protein content of the supernatant was determined by the Lowry method (Lowry, Rosembroug, Lewis, & Randall, 1951) using BSA as the standard. The protein content of the supernatant at pH 12.0 was defined to correspond to 100% protein solubility. Percent of PS at pH values in the range 3.0–10.0 was expressed as (protein content of the supernatant at a tested pH) × 100/(protein content of the supernatant at pH 12.0). All determinations were conducted in triplicates.

2.9. Emulsion preparation

The three pea protein solutions with a constant protein concentration of 1.0% (w/v) were prepared with pH adjusted to desired pHs (3.0–9.0), and stirred using a magnetic stirrer for 2 h at room temperature, and then stored overnight at 4 °C to allow complete hydration. Sodium azide (0.02%, w/v) was used as an antimicrobial agent. Each protein solution or dispersion was mixed with soy oil at oil fraction ($ø$) = 0.1, and pre-homogenized using the high-speed dispersing and emulsifying unit (model IKA-ULTRA-TURRAX® T25 basic, IKA® Works, Inc., Wilmington, NC) at 15,000 rpm for 1 min. Then, the pre-homogenized dispersions were further homogenized though a Microfluidizer (M110EH model, Microfluidics International Corporation, Newton, MA) for one pass at a pressure level of 40 MPa. The fresh emulsions were subject to analysis, or stored at room temperature for various periods of time (e.g., 24 h) prior to further analysis.

2.10. Evaluation of emulsion characteristics

2.10.1. Droplet-size distribution and volume-average droplet size ($d_{4,3}$)

Droplet-size distribution of various freshly prepared or stored (24 h) emulsions were determined using a Malvern MasterSizer 2000 (Malvern Instruments Ltd, Malvern, Worcestershire, UK). Deionized water, or 1.0% (w/v) SDS solution was used as the dispersant. The relative refractive index of emulsion was taken as 1.095, that is, the ratio of the refractive index of soy oil (1.456) to that of the continuous phase (1.33). Droplet-size measurements are reported as volume-average droplet size, $d_{4,3}$ ($=\sum n_i d_i^4/\sum n_i d_i^3$) and surface-average diameter, $d_{3,2}$ ($=\sum n_i d_i^3/\sum n_i d_i^2$, where n_i is the number of droplets with diameter d_i). All determinations were conducted at least in duplicate.

2.10.2. Flocculation and coalescence indices (FI and CI)

FI of freshly prepared (0 h) or stored (after 24 h) emulsions was evaluated according to the method described by Castellani, Belhomme, David-Briand, Guérin-Dubiard, and Anton (2006). The emulsions were diluted in deionized water with and without 1% (w/v) SDS, and the $d_{4,3}$ of droplets was determined as the above. The percentage of FI was calculated as follows:

$$\mathrm{FI}(\%) = \left[(d_{4,3}\ \text{in water})/(d_{4,3}\ \text{in 1\% SDS}) - 1.0\right] \times 100$$

CI (%) of the emulsions after 24 h of quiescent storage was evaluated according to the method of Palazolo, Sorgentini, and Wagner (2005) as follows:

$$\mathrm{CI}(\%) = \left(d_{4,3(24\mathrm{h})}/d_{4,3(0\mathrm{h})} - 1\right) \times 100$$

where $d_{4,3\ (24\mathrm{h})}$ and $d_{4,3\ (0\mathrm{h})}$ were the $d_{4,3}$ of droplets in the emulsions, freshly prepared (0 h) or after storage of 24 h, respectively. For this experiment, all the emulsions were determined with 1% SDS as the diluent.

2.10.3. Creaming index

Creaming index (%) of various emulsions was determined as described by Firebaugh and Daubert (2005), with some modifications. Each emulsion (10 mL) was filled into a glass test tube (1.5-cm internal diameter × 12-cm height) and then stored at ambient temperature (placed in a perpendicular state). Height of serum (H_s) and total height of emulsion (H_t) were recorded after quiescent storage up to 7 days at ambient temperature for different period times. Mean and standard deviations of three replicates were reported. Percentage of creaming index was reported as (H_s/H_t) × 100.

2.10.4. Percentage of adsorbed proteins (AP%) and interfacial protein concentration (Γ)

AP% and Γ of various fresh emulsions (above) were determined using the method described by Ye (2008) and Puppo et al. (2011), with a few modifications. In brief, each fresh emulsion (1 mL) was centrifuged at 10,000 *g* for 45 min at room temperature. After the centrifugation, two phases were observed: cream layer (or concentrated oil droplets) at the top of the tube and aqueous phase of the emulsion at the bottom. The cream layer was carefully removed using a syringe, and the subnatant was filtered through a 0.22 μm filter (Millipore Corp.). Protein concentration of the filtrate (C_f) was determined with the Lowry method using BSA as the standard. Initial protein solution was also centrifuged at the same conditions to allow determination of protein concentration (C_s) in the supernatant. The Γ (mg/m^2) was calculated as:

$$\Gamma\left(\mathrm{mg/m^2}\right) = \left(C_s - C_f\right)^{*} d_{3,2}/6ø$$

The AP% was calculated as $(C_s - C_f)^*100/C_o$, where C_o is the initial protein concentration of the protein solutions applied for the emulsion preparation (in the present work, it was 10.0 mg per mL of solution).

2.11. Statistics

An analysis of variance (ANOVA) of the data was performed, and a least significant difference (LSD) with a confidence interval of 95% was used to compare the means.

3. Results and discussion

3.1. Polypeptide composition

Fig. 2 shows the SDS-PAGE profiles of polypeptide constituents in the three pea proteins (PPI, PL and PV), under reducing and nonreducing conditions. Reducing electrophoresis of PPI showed bands with molecular weight (MW) ranging from about 97 to 10 kDa (Fig. 2, Lane 1, left), and most of bands could be attributed to the polypeptide constituents of legumin and vicilin. Legumin, a hexameric protein, dissociates into two kinds of subunit peptides (α; acidic 38–40 kDa and β, 19–22 kDa), when the electrophoresis was performed under reducing conditions (Casey, 1979; Gueguen, Barbot, & Schaeffer, 1988; Shewry, Napier, & Tatham, 1995). Vicilin is a trimeric protein, composed of three heterogenous polypeptide subunits. The major polypeptide subunits with MW of 70, 50 and 30–35 kDa have been reported, with minor components of lower MW (<19 kDa; Gatehouse, Lycet, Croy, & Boulter, 1982; Koyoro & Powers, 1987). The 70 kDa-polypeptide has been actually recognized to be convicilin, a third storage protein in peas (Barac et al., 2010; Koyoro & Powers, 1987), though in some cases, it was denoted as α-subunit of vicilin (O'Kane, Happe, Vereijken, Gruppen, & van Boekel, 2004b). According to densitometric analysis (on the reducing lane of PPI; Fig. 2, Lane 1), the relative contents of legumin and vicilin (including convicilin) were approximately estimated to be 53 and 39%, respectively (Data not shown). PPI also contained lipoxygenase with MW of about 97 kDa (Fig. 2, Lane 1), with a relative content of about 7.2% (relative to total protein content). The SDS-PAGE profile of PPI is basically similar to that in the previous literature (Barac et al., 2010; Shand et al., 2007; Sun & Arntfield, 2010; Taherian et al., 2011).

As expected, legumin consisting of α and β polypeptides is the prominent component in the PL. In this sample, the polypeptide constituents of vicilin and convicilin could be still observed, but the relative ratios of these polypeptides to those of legumin was much less as compared with PPI (20.5–79.5 vs 39–53) (Fig. 2, Lanes 1 and 2), indicating efficient concentration of legumin in the PL. Similarly, the polypeptide subunits (e.g., 50 and 30–35 kDa) of vicilin were considerably enriched in PV (Fig. 2, Lane 3), in which the relative content of vicilin (and convicilin) was increased to about 65%. The purity of legumin (79.5%) in the PL is higher than that (68.5%) reported by Klassen and Nickerson (2012), but the purity of vicilin in the PV was much lower (65% vs 90.9%). The differences are reasonable, since the cross-contamination of legumin, vicilin and convicilin seems to be common (Koyoro & Powers, 1987; O'Kane et al., 2004a).

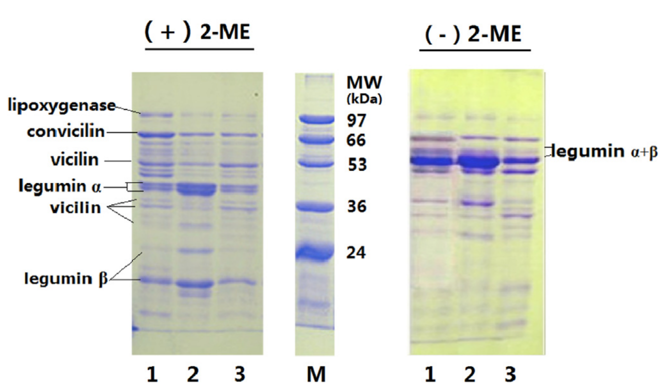

Fig. 2. SDS-PAGE profiles of pea proteins under reducing and nonreducing conditions. Lanes (1–3) represent pea protein isolate (PPI), partially purified legumin and vicilin (PL and PV), respectively. M: protein markers.

3.2. Free SH and SS contents

In the legumin molecules, each acidic (α) polypeptide is usually associated with one basic (β) polypeptide through one disulfide bonding, while the vicilin is devoid of disulfide bonds. As expected, the PL contained total free SH and SS contents of 21.8 and 9.1 μmol/g, significantly higher than those (14.9 and 5.7 μmol/g) for the PV, respectively (Table 1). The differences in free SH and SS contents are basically consistent with the differences in amino acid composition between legumin and vicilin (Koyoro & Powers, 1987), where it was indicated that the relative contents of cystine and methionine residues for the legumin were distinctly higher than that of vicilin. In both the globulins, the SS contents were much less than the free SH contents (Table 1), indicating that most cysteine residues existed as free SH groups rather than as disulfide linkages. This pattern of SH and SS contents is distinctly different from that observed for buckwheat proteins with 11S globulin as the major component (Tang, Wang, Liu, & Wang, 2009). Actually, the presence of SS bonds in the PV could be largely attributed to the contamination of legumin components, e.g. with a relative content of 35.4% (Fig. 2, Lane 3).

The SS content of PPI was expectedly in the midst of KL and KV (Table 1), further indicating that the SS content was contributed by the legumin component. By contrast, the total free SH content or relative ratio of exposed SH groups (relative to total SH groups) of PPI was even significantly higher than that of PL. The higher total free SH groups in the PPI might reflect the fact that legume storage proteins also contain a certain amount of albumins (e.g., protease inhibitors) that are rich in cysteine (Papalamprou, Doxastakis, & Kiosseoglou, 2010). O'Kane et al. (2005) also observed that the protein isolates from peas with different cultivars exhibited higher contents of free SH groups and total amount of cysteine residues than the purified legumins.

3.3. Surface hydrophobicity (H_o)

H_o of a protein is an important physicochemical parameter affecting its surface-related functional properties. The H_o of PL at

Table 1
Free SH and SS contents, H_o, DSC characteristics of pea protein isolate (PPI) and partially purified legumin and vicilin (PL and PV). All the data are means and standard deviations of duplicate measurements.

Samples	Free SH content (μmol/g)		SS content (μmol/g)	H_o	DSC characteristics	
	Total	Exposed			T_d (°C)	ΔH (J/g)
PPI	24.8^{C}	9.52^{B}	7.66^{B}	2734^{C}	82.1^{A} (103)[a]	7.0^{B}
PV	14.9^{A}	3.51^{A}	5.70^{A}	1005^{A}	83.2^{A}	3.8^{A}
PL	21.8^{B}	3.72^{A}	9.05^{C}	2075^{B}	92.3^{B}	12.0^{C}

T_d: the peak temperature of thermal transitions of pea proteins; ΔH: combined enthalpy change of thermal transitions. Different superscripts (A–C) represent significant difference at $p < 0.05$ level among different samples.
[a] A second peak.

pH 7.0 was almost 2 times that of PV (2075 vs 1005; Table 1), indicating much higher exposure of hydrophobic clusters. A similar result about the H_o of pea legumin and vicilin has been previously reported by Koyoro and Powers (1987), though a different fluorescence probe (*cis*-parinaric acid) was used. The difference might be caused by variables in their amino acid or constituent composition, and susceptibility of conformations to processing stresses. Like α, α'-subunits of soybean's vicilin-like protein, β-conglycinin, the subunits of vicilin and convicilin have a core region (and a highly charged N-terminal extension region) (O'Kane et al., 2004a). Thus, the hydrophilic nature of these subunits might largely account for the decreased H_o of PV.

The H_o (2734) of PPI was even markedly higher than that of PL (Table 1). This observation is consistent with that for PPIs prepared by isoelectric precipitation and salt extraction (Karaca et al., 2011), where the H_o of the former PPI was significantly higher than that of the latter. In this previous work, the authors largely attributed the higher H_o to the higher globulin content in the sample, based on the rationales, 1) H_o of globulins is higher than albumins; 2) the isoelectric-precipitated isolate contained higher globulin content than that by salt extraction (Papalamprou et al., 2010). The present observation distinctly differs from this explanation, since PPI contained a certain amount of albumins (though the albumins were almost absent in PL; size exclusion chromatography data not shown) but exhibited higher H_o. Thus, it is probably because the higher H_o for PPI (relative to PL) might be associated with protein unfolding and subsequent exposure of hydrophobic clusters (initially buried within the molecules), caused during the alkaline extraction and acid precipitation.

3.4. DSC characteristics

In the partially purified globulins PR and PL, only an endothermic peak with T_d of 83.2 and 92.3 °C was observed (Table 1), clearly attributed to the denaturation of vicilin and legumin components, respectively. The T_d data are consistent with the results observed for vicilin (or legumin)-like soy β-conglycinin and glycinin (Renkema, Lakemond, de Jongh, Gruppen, & van Vliet, 2000). The difference in T_d between PR and PL has been generally recognized to be associated with the difference in SS content between these two globulins, since the protein with more SS contents exhibits higher thermal stability (Kinsella, 1982). The ΔH, indicative of extent of ordered structure in the sample (Arntfield & Murray, 1981), of PL was considerably higher than that of PV (12 vs 3.8 J/g; Table 1). This observation is in agreement with the differences in polypeptide or protein heterogeneity of these two samples (Fig. 2), suggesting that the proteins in PL were much more ordered in structure than those in PV. The T_d and ΔH for PL are well consistent with that reported for salt-extracted PPI (Sun & Arntfield, 2011a, 2011b), where only one endothermic peak with T_d and ΔH ranging from 90.5 to 93.3 °C and from 8.3 to 16.8 J/g, respectively, was observed.

In the DSC thermogram of PPI, two individual thermal transitions with T_d of 82.1 (minor) and 103 °C (major) were observed (Table 1). As expected, the ΔH (7.0 J/g) was lower than that of PL (12.0 J/g), but higher than that of PV (3.8 J/g) (Table 1). The T_d of the minor peak is basically consistent with that of PV, but that of the major peak was considerably higher than that observed for PL (Table 1). One reasonable explanation is that the conformation of the vicilin component was little affected during the preparation process, while that of the legumin component were much more susceptible. Thus, the increased T_d for the legumin component might reflect enhanced intermolecular protein–protein interactions in PPI (as compared to the PL). The T_d and ΔH data for the two transitions are considerably higher than that (67 and 85 °C, and <1.0 J/g) previously reported for native PPI (Shand et al., 2007), though in this previous work, the DSC was conducted at a higher scanning rate (10 °C/min as compared to 5 °C/min in the present work). The reason for causing this so considerable difference seems to be difficult to explain, since physicochemical properties of the obtained PPI were not characterized in the previous work.

3.5. pH-dependent PS profiles

Solubility is an important prerequisite for a protein to be an effective functional ingredient in a number of foods, e.g., emulsions, foams and gels, since it may affect texture, color and sensory attributes of products. PS is related to balanced protein–protein and protein–solvent interactions. The PS of proteins is markedly affected by pH, ionic strength, ion types, temperature, solvent polarity, and processing conditions (Damodaran, 1996). Fig. 3 shows the PS of the three pea proteins (PPI, PV and PL) at various pH values ranging from 2.0 to 10.0. We can globally see that all the samples similarly exhibited typical pH-dependent PS profiles of legume proteins, with the PS minimal at pH 5.0–6.0 and progressively increasing as the pH deviated above 6.0 or below 5.0. Similar pH-dependent PS profiles of pea proteins have been observed for commercial or native PPI (Adebiyi & Aluko, 2011; Fernández-Quintela, Macarulla, del Barrio, & Martinez, 1997; Shand et al., 2007; Taherian et al., 2011; Tian et al., 1999), pea protein concentrate (Boye et al., 2010; Tian et al., 1999), vicilin (Casanova, Nakamura, Masuda, Lima, & Fialho, 2008; Koyoro & Powers, 1987; Pedrosa et al., 1997) and legumin (Koyoro & Powers, 1987).

However, it can be still observed that the PS varied with the type of pea proteins and pH. At pH 7.0 or above, the PS of PL was identical to that of PR, but higher than that of PPI (Fig. 3). This slightly differs from the result reported by Karaca et al. (2011) that the solubility at 7.0 of PPI by isoelectric precipitation was considerably higher than that by salt extraction. On the other hand, at pH below 5.0, the PS progressively decreased in the order: PL > PV > PPI. Interestingly, the PS at pH 3.0 or below of PL was even higher than that at pH 7.0–9.0. The observations for PL and PV are inconsistent with that reported for purified legumin and vicilin (Koyoro & Powers, 1987), where the PS at pH 3, 4 and 7 of vicilin was considerably higher than that of legumin. These differences in PS between different works seem to be common for plant proteins in the literature. For example, Karaca et al. (2011) indicated that the PS (38–61%) at pH 7.0 of PPI was considerably lower than that (~97%) of SPI, prepared using a same method. This is much deviating from the previous

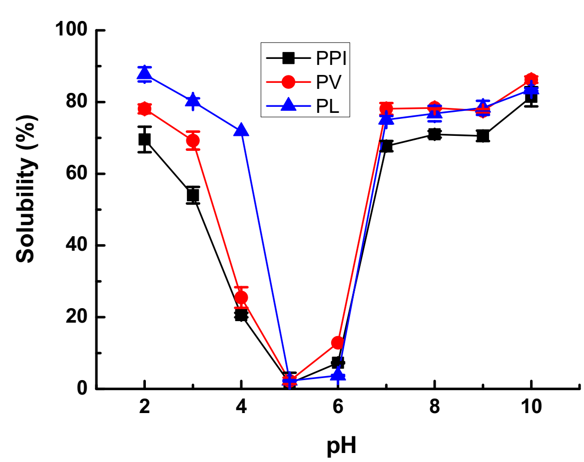

Fig. 3. The pH-dependent PS profiles of PPI, PL and PV. Each data is means and standard deviations of at least triplicate measurements.

viewpoint that PPI exhibits much better solubility than SPI (Aluko et al., 2009; Swanson, 1990).

3.6. Emulsifying properties

3.6.1. Emulsifying ability

Emulsifying properties of various pea proteins (PPI, PL and PV) at pH 3.0–9.0 were evaluated at a protein concentration of 1.0% (w/v) and an oil fraction of 0.1 using droplet-size analysis. Fig. 4 shows typical droplet-size distribution profiles of various fresh emulsions, diluted in 1% SDS or in deionized water. The $d_{4,3}$ of the droplets, diluted in 1% SDS or deionized water, was calculated and summarized in Table 2. The $d_{4,3}$ in 1% SDS can reflect ability of a protein to help dispersion of oil phase into an aqueous medium, since the presence of 1% SDS may inhibit bridging flocculation of oil droplets, thus keeping individual droplets separate in the emulsions. In the presence of 1% SDS, all the emulsions exhibited a prominent distribution peak, but the magnitude and location of the prominent peak considerably varied with the type of proteins and pH (Fig. 4A–D). Overall, we can observe that the $d_{4,3}$ (in 1% SDS) was considerably greatest at pH 5.0 (around the isoelectric point, *p*I), and it progressively decreased as the pH deviated from 5.0 (Table 2). This observation is well in agreement with the pH-dependent PS profiles of the proteins (Fig. 3), confirming the fact

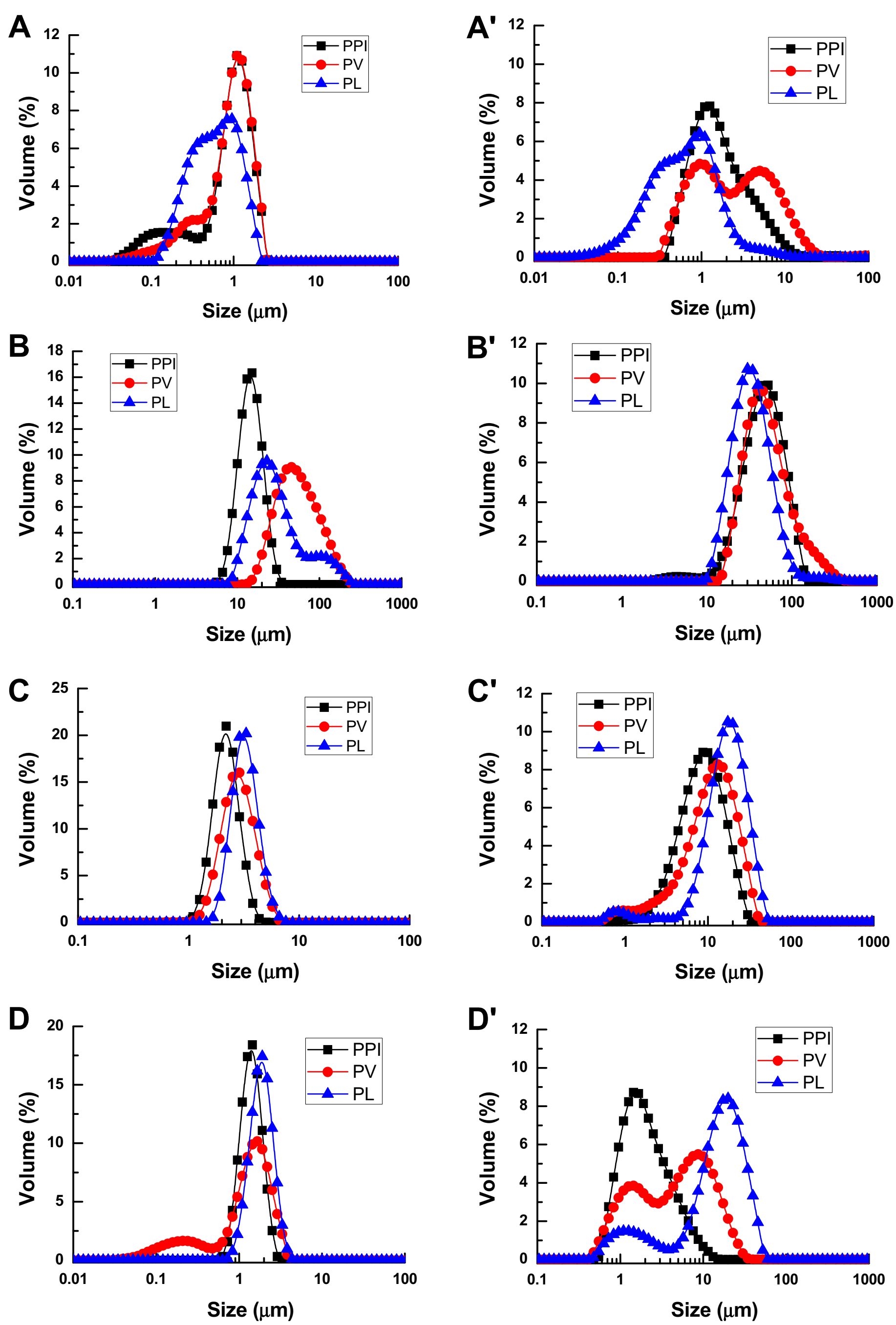

Fig. 4. Typical size distributions of droplets in emulsions stabilized by PPI and partially purified globulins (PL and PV), at various pH values of 3.0 (A, A′), 5.0 (B, B′), 7.0 (C, C′) and 9.0 (D, D′), respectively. A–D: diluted in 1% SDS; A′–D′: diluted in deionized water.

Table 2
Emulsion characteristics, including volume-mean droplet size ($d_{4,3}$), flocculation and coalescence indices (FI and CI), percentage of adsorbed proteins (AP%) and interfacial protein concentration (Γ) of pea protein-stabilized emulsions at pH 3–9, freshly prepared or after a storage of 24 h. All values are means of at least duplicate measurements.

Samples	pH	$d_{4,3}$ (μm)				FI		CI (at 24 h)	Interfacial protein adsorption	
		0 h		24 h		0 h	24 h			
		Water	1% SDS	Water	1% SDS				AP (%)	Γ (mg/m^2)
PPI	3	2.29	1.02	10.2	5.20	2.24	1.95	409.4	19.9	0.92
	5	52.1	16.2	53.9	19.2	3.22	2.80	18.7	1.00	1.20
	7	10.5	2.40	17.9	2.47	4.36	7.26	2.75	11.2	2.10
	9	2.69	1.55	5.34	1.56	1.74	3.42	0.65	12.8	1.53
PV	3	5.22	1.06	6.24	1.08	4.93	5.78	1.89	22.7	1.06
	5	68.0	67.6	97.6	80.5	1.01	1.21	19.1	1.83	7.55
	7	13.6	3.15	20.1	3.17	4.3	6.35	0.63	15.0	3.54
	9	7.02	1.47	9.06	1.48	4.77	6.12	0.68	16.8	0.80
PL	3	0.99	0.75	1.19	0.76	1.32	1.57	1.33	32.9	1.39
	5	44.5	43.8	63.3	54.4	1.07	1.16	24.3	1.69	3.72
	7	19.9	3.58	20.1	3.73	5.56	5.39	4.11	10.3	2.87
	9	18.3	2.06	22.5	2.08	8.85	10.83	0.97	15.4	2.41

that solubility of proteins plays an important role in their emulsifying properties, though 100% solubility is not an absolute requirement (Damodaran, 1997). Similar observations have been reported for pea globulins in terms of emulsion capacity (legumin, vicilin or their mixtures; Koyoro & Powers, 1987), and PPI in terms of EAI (Barac et al., 2010).

Besides the PS, the emulsifying properties of proteins can be affected by many other molecular parameters, including H_0, conformation state, and conformational flexibility (Damodaran, 1997). Although the PS at pH 7.0 and 9.0 was similar for any pea protein (Fig. 3), the $d_{4,3}$ of droplets (in 1% SDS) at pH 9.0 was significantly lower than that at pH 7.0 (Table 2). Thus, the higher emulsifying ability at pH 9.0 (relative to pH 7.0) could be largely attributed to higher extent of pH-induced structural unfolding of protein molecules. Furthermore, the $d_{4,3}$ order (in 1% SDS) at pH 7.0 for various emulsions was basically the same as that of the H_0 for the proteins (Tables 1 and 2), confirming positive correlation between H_0 and emulsifying properties of the proteins (Damodaran, 1997).

Interestingly, for any pea protein, the $d_{4,3}$ at pH 3.0 was least among all the test pH values, which was even significantly lower than that at pH 9.0 (Table 2), indicating that pea proteins exhibited much better emulsifying ability at pH below *pI* (e.g., pH 3.0), than at neutral or alkali pH (e.g., pH 9.0). This is basically in accordance with the observation of Gharsallaoui et al. (2009) that pea protein-stabilized emulsions at acidic pH (e.g. pH 2.4) had more homogenous droplet-size distributions and stable against creaming. They proposed that at acidic conditions, pea protein forms a stronger and denser viscoelastic network when adsorbed at the interface (than at neutral or alkali pH).

On the other hand, we can see that at pH 3.0, the PL exhibited better emulsifying ability than PPI or PV, while at pH 7.0 or 9.0, the situation was the reversal (Table 2). The better emulsifying ability for PL at pH 3.0, similarly observed in the work of Koyoro and Powers (1987), might be associated with its higher PS at this pH (as compared to PV or PPI; Fig. 3). The observation at pH 7.0 is similar to that observed by Dagorn-Scaviner et al. (1987), but different from that by Koyoro and Powers (1987). Dagorn-Scaviner et al. (1987) indicated that vicilin at pH 7.0 is significantly more effective than legumin in terms of its emulsifying capacity. They partly attributed the better emulsifying capacity of vicilin to its faster adsorption rate at the interface.

3.6.2. Flocculated state of oil droplets in fresh emulsions

Fig. 4 shows that the droplet-size distribution (in deionized water) of various fresh emulsions at any pH globally shifted toward larger sizes, as compared with that in 1% SDS, indicating bridging flocculation of oil droplets. The flocculated state of oil droplets was evaluated in terms of flocculation index (FI), as also included in Table 2. The pH dependence of FI for fresh emulsions (at 0 h) varied with the type of pea proteins. For PPI, the FI progressively increased with pH increasing from 3.0 to 7.0, and then considerably declined at pH 9.0 (even much less than that at pH 3.0). For PL or PV, lowest FI was observed at pH 5.0, and FI progressively increased as the pH deviated from 5.0 (Table 2).

At pH 5.0, the FI (3.22) of the fresh PPI emulsion was much higher than that (1.01–1.07) of the PV or PL counterparts (Table 2). In the latter two emulsions, a majority of oil droplets were present in non-flocculated state, though the $d_{4,3}$ of droplets (in 1% SDS) was much higher than that of the PPI emulsion. These observations indicated that at pH 5.0, PPI exhibited stronger ability to be adsorbed at the interface, thus showing higher emulsifying ability, but the droplets in the resultant emulsion had much higher tendency to flocculation (as compared to PV or PL). At pH 3.0, the situation was distinctly different. In this case, lowest FI (1.32) was observed for the PL emulsion, and the FI (4.93) was highest for the PV emulsion (Table 2), suggesting that PL was the most appropriate protein (among all the test pea proteins) to be used to form emulsions with homogenous and fine droplets at pH 3.0. At this acidic pH, the droplets in the PV emulsion exhibited higher tendency to associate one another than those in the PPI emulsion, though these two proteins had similar emulsifying ability (Table 2).

At pH 7.0, all the emulsions exhibited a similarly flocculated state of oil droplets, with FI of 4.3–5.56 (Table 2). However, the flocculated state at pH 9.0 considerably varied with the type of pea proteins. At this pH, the FI of the emulsions progressively increased in the order: 1.74 (PPI) < 4.77 (PV) < 8.85 (PL), though these proteins exhibited similar emulsifying ability (as evidenced by similar $d_{4,3}$ in 1% SDS; Table 2).

3.6.3. Flocculation and coalescence stability

The stability of various emulsions formed at various pH values (3.0–9.0) upon storage of 24 h was evaluated in terms of FI and CI. In usual, both flocculation and coalescence of oil droplets in emulsions simultaneously occur during storage. Thus, the mechanism for the emulsion instability might vary with the type of pea proteins and pH. In the PPI case, the FI at pH 7.0 or 9.0 was generally markedly higher for the stored (after 24 h) emulsion than the fresh counterpart, while at pH 3.0 or 5.0, the FI was on the contrary decreased by the storage (Table 2). On the other hand, the CI (at 24 h) progressively and remarkably decreased from 409.4 to 0.65 with increasing pH from 3.0 to 9.0 (Table 2). These observations indicated that for the PPI emulsions, droplet coalescence was the prominent mechanism for the emulsion instability at acidic conditions, while at neutral or alkali conditions, bridging flocculation played the major role. The progressive decrease in CI upon increase in pH (Table 2) to a large extent supported the previous argument that pea proteins form a more viscoelastic film at the interface at more acidic conditions (Gharsallaoui et al., 2009), since the coalescence stability of droplets is closely related to the viscoelasticity of interfacial films (Damodaran, 1997).

The situation in the PV and PL emulsions was distinctly different from that in the PPI emulsion. In these two emulsions, relatively high CI (19.1–24.3) was observed only at pH 5.0, and the CI (0.63–4.11) at other pH values was considerably lower as compared to that at pH 5.0 (Table 2). This is consistent with the fact that good solubility is a preresquisite for proteins to exhibit good emulsifying properties (Damodaran, 1996). By contrast, the FI of these two emulsions at any test pH slightly increased after the storage (Table 2). The observations indicated that these two emulsions was most unstable at pH 5.0, and the underlying mechanism for the instability was by means of droplet coalescence; these emulsions exhibited relatively high stability against flocculation and/or coalescence, at pH deviating from 5.0.

3.6.4. Creaming stability

Any protein-stabilized emulsion is unstable, and if stored for a period long enough, it will develop into two layers, cream (consisting of oil droplets) and serum. Table 3 shows the creaming index of the three pea protein emulsions at various pH values (3.0–9.0), upon quiescent storage up to 7 days. As expected, various emulsions showed different creaming behaviors, depending on the type of pea proteins and the applied pH (Table 3). For any emulsion, the creaming stability at pH 5.0 was least among all test pH values, since these emulsions developed rapidly into two layers, even after a storage period of 1 day. At pH 5.0, the creaming index at 1st day of all the emulsions was basically identical (56.2–62.7%), and it slightly increased as the storage period was prolonged up to 7 days (Table 3). In usual, the creaming rate is affected by many variables, e.g., droplet size, flocculated state of oil droplets, stability against flocculation and/or coalescence. Thus, the creaming instability at pH 5.0 could be largely attributed to large droplet sizes and decreased stability against coalescence.

In the PPI case, the creaming behavior of the emulsion at pH 7.0 was similar to that at pH 5.0 (Table 3). Furthermore, although the creaming index (21.4%) at 1st day at pH 3.0 was much less than that (57.1%) at pH 7.0, the creaming index at pH 3.0 and 7.0 was almost identical after storage of 3 days or above (Table 3). Together with droplet-size data (in both deionized water and 1% SDS; Table 2), these observations imply that the droplet size and flocculated state might be not the sole parameter affecting the creaming behavior of the PPI emulsions. On the other hand, we can see that the PPI emulsion at pH 7.0 exhibited much less stability against flocculation than that at pH 5.0; the droplets in the emulsion at pH 3.0 had much higher tendency to coalescence as compared to those at pH 7.0 (Table 2). Thus, besides the droplet size and flocculated state, the flocculation and/coalescence stability upon storage might also contribute to the creaming behavior of the PPI emulsions. This can be further corroborated by the observation that the creaming stability of the PPI emulsion at pH 9.0 was remarkably higher than that at pH 7.0 (Table 3). The enhanced creaming stability at pH 9.0 was clearly associated with the much less droplet size and FI of the emulsions (fresh or after storage) at pH 9.0.

Interestingly, there was no creaming occurring in the PV emulsions at pH 7.0 or 9.0, and in the PL emulsions at pH 3.0 or 9.0, upon storage up to 7 days (Table 3), indicating that these emulsions exhibited excellent stability against creaming. By contrast, the creaming stability of the PV emulsion at pH 3.0, or the PL emulsion at pH 7.0 was relatively lower; and in these cases, distinct creaming occurred after storage of 3 days, though the creaming index (16.7–18.7%) was much less than that at pH 5.0 (Table 3). These observations are basically consistent with the FI and CI data of these emulsions (Table 2), suggesting that the creaming behavior of these emulsions at pH deviating from 5.0 was much more related to the flocculation and/or coalescence stability, than associated with the droplet size or flocculated state in the emulsions.

3.6.5. Interfacial protein adsorption

AP% and Γ data of various fresh emulsions at various pH values (3.0, 5.0, 7.0 and 9.0) are included in Table 2. For any pea protein emulsion, the AP% (1.0–1.83%) was least at pH 5.0 (Table 2), signifying that most of the proteins in the emulsions remained unadsorbed. This observation agrees with the fact that the emulsifying ability of these proteins was lowest at this pH, mainly due to protein insolubility (Table 2). As the pH deviated from 5.0, the AP% considerably increased. Overall, we can observe that 1) for any kind of emulsions, the AP% at pH 3.0 was significantly higher than that at pH 7.0 or 9.0; 2) at pH 3.0, the AP% of the PL emulsion was considerably higher than that of the PPI or PV emulsions, while at pH 7.0 (or 9.0), highest AP% was observed for the PV emulsion. The AP% data at pH 3.0 for these emulsions are well in accordance with the PS data at the same pH (Fig. 3), suggesting that the protein adsorption at the interface pH 3.0 was closely associated with the PS.

Besides the PS, the proportion of protein adsorption is also affected by other parameters, e.g., surface hydrophobicity/charge, protein–protein interactions, and conformational flexibility of the proteins at the interface. This can be corroborated by the consistency between the PS and AP% at pH 3.0 and 7.0. For example, the PS at pH 3.0 of all the pea proteins was comparable to or lower than that at pH 7.0, but the AP% at pH 3.0 was much higher (Fig. 3 and Table 2). In this situation, the higher repulsion interaction between the proteins, as well as more compacted conformations at pH 7.0 (relative to pH 3.0) might largely account for the lower AP% at pH 7.0.

The Γ of the emulsions considerably varied with the type of pea proteins and the applied pH. There was no distinct relationship between the AP% and Γ data (Table 2). This differs from the observation for soy β-conglycinin and glycinin emulsions (Puppo et al., 2011), where the Γ followed almost the same behavior as that of AP%. Although the AP% was lowest at pH 5.0 for any pea protein emulsion (among all the test pH values), the Γ was not least at this pH. On the contrary, the Γ at pH 5.0 for the PV and PL emulsions was highest among all the test pH values (Table 2). This observation reflects that during the emulsification, insoluble

Table 3
Creaming index of various pea protein emulsions formed at various pH values (3.0–9.0), upon storage up to 7 days. Each data is means of at least duplicate measurements.

Emulsions		Creaming index (%)		
		1 Day	3 Days	7 Days
PPI	pH 3	21.4	57.1	60.0
	pH 5	61.1	62.5	62.5
	pH 7	57.1	58.6	60.0
	pH 9	0	0	30.7
PV	pH 3	0	16.7	18.1
	pH 5	56.2	57.5	61.6
	pH 7	0	0	0
	pH 9	0	0	0
PL	pH 3	0	0	0
	pH 5	62.7	64.0	65.3
	pH 7	0	18.7	21.3
	pH 9	0	0	0

proteins can be also adsorbed at the interface, though the AP% was considerably lower. Once the insoluble proteins are adsorbed, they will form much thicker interfacial film at the adsorption sites. On the other hand, we can still observe that except at pH 5.0, all the Γ data were basically consistent with the $d_{4,3}$ data (in 1% SDS) (Table 2), indicating a close relationship between the emulsifying ability and the Γ.

In general, the thicker the adsorbed protein layer, the lower is the coalescence rate (or the higher is the coalescence stability). Besides the film thickness, the coalescence stability is determined by viscoelastic properties and/or integrity of interfacial films (Damodaran, 1997). The relatively higher CI at pH 5.0 (close to the isoelectric point of the protein; Table 2) could thus be attributed to the bad integrity of the protein film, since the protein aggregates or precipitates in the adsorbed layer would create "holes" in the film, causing coalescence. A similar phenomenon has been previously observed for β-lactoglobulin emulsions (Das & Kinsella, 1989). Clearly, the extraordinary instability against coalescence for the PPI emulsion at pH 3.0 (Table 2) cannot be explained in terms of Γ, though the Γ of the PPI emulsions at pH 3.0 was lower than that at other pH values. In this case, we can see that after storage of 24 h, the $d_{4,3}$ (in water) of the PPI emulsion at pH 3.0 remarkably increased by more than 3.4-folds (from 2.3 to 10.2 μm), with increasing extent much greater than in the other emulsions (Table 2). This observation implies that the extraordinary instability might be largely facilitated by flocculation of oil droplets, though the flocculated state at 24 h was even less than that at 0 h.

4. Conclusions

The emulsifying properties of pea proteins considerably varied with the protein composition and pH (3.0–9.0). In general, various pea proteins exhibited least emulsifying ability at pH 5.0, and their emulsifying ability was even better at acidic conditions (e.g., pH 3.0) than at neutral or alkali pH (e.g., pH 9.0), suggesting that these proteins exhibited good potential to be used for the emulsion formulation at acidic conditions. In this aspect, PL (legumin-rich pea proteins) seemed to be more appropriate. On the other hand, the emulsion stability against flocculation, coalescence and even creaming also to different extents depended on the type of the proteins and pH. Interestingly, the PL and PV (extracted at neutral pH) exhibited much better creaming stability than PPI (with alkali extraction), at pH deviating from the *p*I. The emulsifying properties of these proteins were not only related to their PS and H_0, but also associated with the protein adsorption at the interface and the nature (e.g., viscoelasticity) of interfacial protein films. These findings would greatly extend the knowledge about the emulsifying properties of pea proteins, thus providing usable information on characteristics of pea protein-stabilized emulsion formulations.

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